Computer Systems

4th September 2024

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

a) Select ONE (1) option that represents a collection of computer peripherals. **1**

i) CPU, RAM, Motherboard

ii) Keyboard, Mouse, Monitor

iii) Operating System, Application Software, Utility Software

iv) Headphones, Mouse, Internet

b) Select ONE (1) option that is a function of an Operating System. **1**

i) Protecting the user from spyware

ii) Running application software

iii) Designing user interfaces

iv) Compilation of code

c) Structured Query Language (SQL) is a language for manipulating and querying databases. Select ONE (1) option that is **NOT** a valid SQL statement. **1**

i) SELECT

ii) AMEND

iii) UPDATE

iv) DELETE

d) Which type of memory is considered volatile? Select ONE (1) option. **1**

i) ROM (Read-Only Memory)

ii) Flash Memory

iii) RAM (Random Access Memory)

iv) SSD (Solid State Drive)

e) What is the binary representation of the hexadecimal value CF. Select ONE (1) option. **1**

i) 00111010

ii) 10100011

iii) 00110101

iv) 11001111

f) Identify FIVE (5) key components that make up a Central Processing Unit (CPU). **5**

i) Arithmetic Logic Unit (ALU)
ii) Control Unit (CU)
iii) Register File
iv) Graphics Processing Unit (GPU)
v) Memory Management Unit (MMU)
vi) Floating Point Unit (FPU)
vii) Instruction Decoder
viii) Cache Memory

Total 10 Marks

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Question 2

You have been asked to build a Personal Computer (PC) for a close friend. Your friend would like to use the PC for word processing and browsing the internet. They already have a keyboard, mouse, and a monitor.

a) Name FIVE (5) *main computing hardware* components you would need to create a functioning PC? **5**

b) Your friend has informed you that the computer struggles to play games on the PC. A quick diagnosis shows the screen stuttering when attempting to play visually demanding games. **1**

Identify the internal hardware component that you would suggest they upgrade, to ensure visually demanding games run smoothly on the PC?

c) Which component(s) of the PC would require updating if you started to run out of space for permanently storing documents? **1**

d) The Central Processing Unit (CPU) contains a small amount of cache memory. Explain the purpose of cache memory. **3**

Total 10 Marks

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Question 3

a) Computer systems process binary numbers which can be expressed in different formats.

i) Add binary numbers 10101 and 00100. Give the result as a binary number. **1**

ii) Convert the numbers from i) to decimal to prove your answer. Show your working. **3**

iii) Convert the hexadecimal number F8D4 to its decimal value. Show your working. **2**

iv) Use two's complement method to negate the binary number 11010011. **2**

v) What is the octal equivalent of the hexadecimal value B35? Show your working. **2**

Total 10 Marks

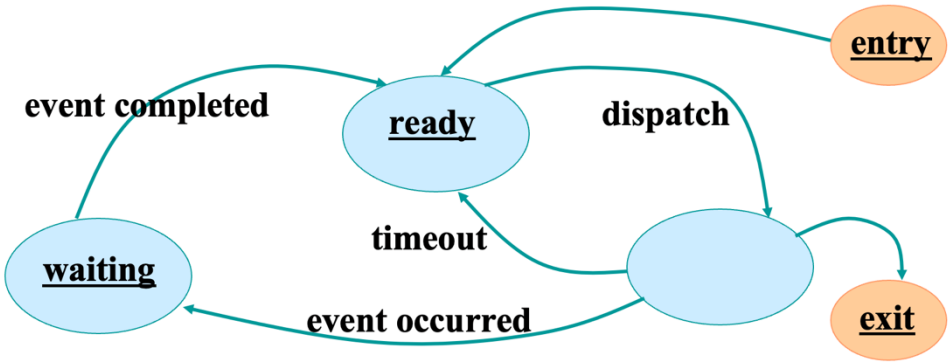
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Question 4

The Operating System (OS) is a crucial part of any computer system. It serves as the interface between the user and the components.

a) In addition to the OS providing a user interface, outline and describe THREE (3) additional purposes of the OS? **9**

b) Identify the missing state as part of the process state diagram: **1**



Total 10 Marks

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Question 5
The Central Processing Unit (CPU) is the fundamental digital data processor in a computer system.

a) Identify the THREE (3) main functions of a CPU? **3**

b) Explain the purpose of an operand in a CPU? **2**

c) Explain the purpose of an opcode in a CPU? Give ONE (1) example of an operation it performs. **3**

d) Before the CPU processes any data, it is recommended that data is checked via a parity checker. Outline how a parity checker works. **2**

Total 10 Marks

Question 6
Data communications networking is an integral part of modern computer systems.

a) Identify THREE (3) main goals of successful networking of data? **3**

b) The internet is an example of a mode of data transmission. Consider the following statement: **3**

"The internet is based on a synchronous transmission system."

State whether the above statement is *true* or *false* and explain how a synchronous mode of transmission functions.

c) BUS and RING are TWO (2) examples of LAN topologies. Name ONE (1) advantage and ONE (1) disadvantage of each topology. **4**

Total 10 Marks

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Question 7

a) Micro-controllers are widely used in a number of different areas. Provide FOUR (4) examples of application areas where micro-controllers are used. **4**

b) Explain the difference between a polled interrupt and a vectored interrupt. **2**

c) Real Time Operating Systems (RTOS) are specialised operating systems designed to handle real-time applications with precise timing and high reliability.

Outline FOUR (4) distinct characteristics of a RTOS's operation that you may find on an embedded system, such as a car's electronic control unit (ECU). **4**

Total 10 Marks

Question 8

Reduced Instruction Set Computing (RISC) is a CPU architecture design philosophy that emphasises simplicity and efficiency of operation.

a) Identify THREE (3) features of RISC that allow it to operate simply and efficiently. **3**

b) Identify THREE (3) advantages of RISC when compared to Complex Instruction Set Computing (CISC). **3**

c) Explain the role of context switching when TWO (2) programs compete for the CPU time. **4**

Total 10 Marks

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Question 9

a) Scenario

A university library database keeps track of its books, members and the borrowing process. The structure of the database is as follows:

1. **Books:** Each book has a unique ISBN, a title, an author and a publication year.

2. **Members:** Each member has a unique member ID, a name, an email address and a membership start date.

3. **Borrowing:** Each borrowing transaction records the member who borrowed the book, the book borrowed, the date borrowed and the due date.

i) In relation to the given scenario, identify THREE (3) attributes that exist in the university library database. **3**

ii) In relation to the given scenario, what Structured Query Language (SQL) command would be used to add the following information to the *Books* table? **1**

9780763707996, Computer Science Illuminated, Nell B. Dale, 2004

b) State the primary purpose of Domain Name Service (DNS) in computer networking. **1**

c) Some developers use virtual machines. Describe what a virtual machine is and outline THREE (3) of its purposes. **5**

Total 10 Marks

Question 10

To enhance a computer system's performance, advanced technologies have been developed, besides increasing CPU clock speeds.

a) A CPU, with its pipeline switched off, executes 1800 instructions in 5400 clock cycles. What is the average clock cycle per instruction executed? **1**

b) Explain the purpose of a CPU pipeline? **2**

c) Explain how a multi-core CPU has enhanced performance over a single-core CPU. **2**

d) Outline FIVE (5) characteristics of distributed computing? **5**

Total 10 Marks

End of paper

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